

Shubham Acharya

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Professional Summary

AI Engineer / Python Backend Developer with hands-on experience building Generative AI applications, agentic workflows, and RAG-based systems using Python, FastAPI, LangChain, and LangGraph. Experienced in designing AI workflow orchestration, tool-calling systems, vector search pipelines, structured LLM outputs, and evidence-grounded AI applications. Proficient in PostgreSQL, Pandas, ChromaDB, Docker, Google Cloud, Vertex AI, OAuth/JWT, and LLM evaluation. Strong focus on building modular, scalable backend systems that combine deterministic processing with LLM-based reasoning.

Technical Skills

Languages: Python, SQL

AI & Generative AI: LangChain, LangGraph, RAG, AI Agents, Multi-Agent Systems, Prompt Engineering, Tool Calling, Vector Search, Embeddings, NLP, Hugging Face, MCP

LLM Evaluation: LangSmith, DeepEval, LLM Evaluation, RAG Evaluation, Evaluation Metrics

Backend: FastAPI, REST APIs, SQLAlchemy, Pydantic, OAuth 2.0, JWT

Data Libraries: Pandas, NumPy

Databases & Storage: PostgreSQL, MySQL, MongoDB, ChromaDB, Supabase

LLMs & AI Providers: Google Gemini, Vertex AI, GPT, Claude, Groq, Llama

Cloud & DevOps: Docker, Docker Compose, Google Cloud Platform, Cloud Run, Vercel, Render

Tools: Git, GitHub

Projects

NexusAI — AI Workflow Automation & Research Platform

 GitHub

Python • FastAPI • LangGraph • LangChain • Gemini • PostgreSQL • Docker

Built a visual node-based AI workflow platform for multi-step research and knowledge pipelines. Designed a modular plugin architecture using metadata and Pydantic schemas, enabling new AI nodes without modifying the core engine. Developed a dynamic LangGraph engine that converts frontend workflow graphs into executable StateGraph workflows and resolves node dependencies.

PrepMate AI — AI-Powered Interview Preparation Platform

 GitHub

LangGraph • LangChain • FastAPI • Vertex AI • PostgreSQL • Docker

Built an AI interview preparation platform with resume analysis, ATS scoring, mock interviews, job search, and career roadmaps. Designed LangGraph workflows for resume evaluation, question generation, candidate assessment, and career planning. Implemented secure FastAPI services with Google OAuth, JWT, PostgreSQL, Supabase, Docker, and Google Cloud.

Medical AI Assistant — Retrieval-Augmented AI System

 GitHub

LangGraph • LangChain • FastAPI • ChromaDB

Developed a RAG-based medical information assistant with intelligent tool routing and evidence-backed responses. Integrated PubMed, OpenFDA, Tavily Search, and Wikipedia. Built PDF ingestion with semantic chunking, embeddings, ChromaDB vector search, and citation-aware response generation.

Data Visualization — Enterprise AI Strategy Intelligence Platform

 GitHub

Python • FastAPI • LangChain • Groq Llama 3.3 • PostgreSQL • ChromaDB • Next.js

Built an enterprise AI platform combining business analytics, document RAG, and external research for evidence-grounded strategy recommendations. Designed SupervisorAgent-based intent classification and execution planning across sales, customer, knowledge, and research tools. Implemented an EvidencePackage layer and deterministic Pandas analytics for revenue, profit, margins, churn, CSAT, and customer risk.

Experience

Digital Computer Instructor

Jul 2025 – Mar 2026

BR Birla Group of Schools, Jodhpur

Conducted hands-on Python, robotics, and electronics sessions through project-based learning. Worked with Arduino, Raspberry Pi, sensors, servo motors, and electronic components to develop robotics projects. Guided students in Python programming, circuit design, hardware integration, and technical problem solving.

Education

Bachelor of Computer Applications (BCA)

2022 – 2025

Aishwarya College, Jodhpur

CGPA: 7.0